Measuring What You Cannot See: A Calibrated Study of Welfare and Equity Erosion in Proxy-Optimizing Incentive Allocation

[Author name and affiliation]

Abstract

Organizations increasingly delegate resource-allocation decisions to systems that optimize what they can measure. When the measurable target (employee retention) is only a proxy for the true objective (employee welfare), optimization pressure can silently erode the objective it cannot see. We study this in an open, CPU-only agent-based testbed in which a budget-limited learner allocates five incentive types across a heterogeneous workforce, observing retention but not a latent satisfaction variable, with a tunable "handcuff" channel through which monetary incentives raise retention *without* raising satisfaction. Across 200 seeds we find that the retention-optimizing learner matches a theory-based static allocator on observed retention while degrading latent satisfaction (-0.037 on a $[0, 1]$ scale, 100% seed-sign-stable) and roughly tripling cross-archetype inequality. Through pre-registered falsification tests we show this degradation is **not** an artifact of our assumptions: it persists at zero crowding-out, on a never-left "founder" cohort (ruling out survivorship composition), in 100% of 250 randomized parameter worlds, and—critically —when the attrition hazard and archetype mix are **calibrated to the IBM HR Analytics benchmark**, in which the off-satisfaction "handcuff" retention channel is itself empirically present (stock-option coefficient -0.51 controlling for satisfaction). We frame this as a *distributional* sharpening of multitask principal–agent theory: a retention-optimizer does not merely lower average welfare, it concentrates the loss on specific archetypes, robustly and under calibration. We also test the natural remedy—a sparse audit of the hidden objective folded into the reward—and report an honest *negative* result: the naive-estimator backfire we first observed appears only under strong satisfaction–retention coupling; a precision-aware estimator, *or* the weaker coupling the benchmark actually exhibits, removes it. Audit remedies matter only in a narrow regime. All code, data, the IBM calibration, and a prompt log for AI-assisted implementation are released.

1 Introduction

Learning systems that allocate limited resources—advertising budget, clinical attention, promotions, retention incentives—are typically rewarded on a metric that can be logged. In personnel settings that metric is almost always *retention*: it is cheap to observe and directly tied to cost. The construct the organization actually cares about—employee welfare, or satisfaction—is expensive to observe and is measured, if at all, through infrequent surveys. This gap between a cheap observable proxy and an expensive latent objective is the classic precondition for Goodhart's law [15]: "when a measure becomes a target, it ceases to be a good measure." A large literature establishes that optimizing a proxy can degrade the true objective, in economics [10, 2] and in machine learning [1, 12, 14, 11]. What is far less understood is a practical, deployment-relevant question: if an organization knows its allocator optimizes a proxy and can afford to occasionally *audit* the true objective, does that audit repair the damage? And does the answer depend on how often it audits, or on how it uses what it learns?

We study three research questions in a controlled simulation:

- **RQ1 (Existence & robustness).** Does optimizing an observable retention proxy degrade unobserved satisfaction and its distribution across worker types, and is that degradation robust to modelling choices rather than an artifact of a single hand-set parameter?
- **RQ2 (Harm vs. trapping).** If satisfaction falls, is it because workers are made less satisfied (per-person harm) or because unhappy workers who would have quit are retained (survivorship composition)?
- **RQ3 (Audit remedy).** Does auditing the hidden objective and folding it into the reward repair the degradation—and is the answer governed by audit *frequency* or by how the audit signal is *estimated*?

Contributions.

1. **An open, reproducible testbed** for proxy-driven incentive allocation with a latent welfare variable, heterogeneous worker archetypes, intrinsic-motivation crowding-out, and a tunable unmediated-retention "handcuff" channel—CPU-only, standard-library, deterministic per seed.
2. **Robust, calibrated evidence of *distributional* welfare erosion.** A retention-optimizing learner degrades latent satisfaction and, more sharply, cross-archetype *equity* (roughly $4\times$ the inequality of a static theory baseline) while matching that baseline on observed retention. We show (via a zero-crowding control, a founder-cohort metric, replacement-scheme sensitivity, a 250-world randomized sweep, *and* calibration of the hazard slope and archetype mix to the IBM HR Analytics benchmark) that the effect is not an assumption artifact. We frame it as a distributional sharpening of multitask agency [10]: the loss concentrates on specific worker types, and the "handcuff" channel is empirically present in the benchmark (stock options).
3. **An honest negative result on the audit remedy**, from two pre-registered falsification tests. A naive audit-reward estimator appears to backfire at low audit volume, but this vanishes both under a precision-aware estimator and under the weaker satisfaction–retention coupling the benchmark actually exhibits—so measurement helps at most in a narrow regime. We report in full the finding that *failed* its kill-tests, not only the ones that survived.
4. **Honest scope.** The study is a conditional, hypothesis-generating simulation with unitless parameters; we report where each finding is and is not robust, and outline a remote-only path to empirical calibration.

2 Related Work

Proxy optimization and reward hacking. That optimizing an imperfect proxy degrades the true objective is formalized for reinforcement learning by Karwowski et al. [11] and quantified as reward-model overoptimization by Gao et al. [8]; both study optimization *pressure* on generic objectives. Conceptual treatments of specification gaming [1, 12] and a formal characterization of reward hacking [14] establish the phenomenon. Our contribution is orthogonal: we hold the optimizer fixed and ask whether *measuring* the hidden objective at a chosen frequency and precision repairs the damage.

Personnel economics. The multitask principal–agent model [10] shows that strengthening incentives on a measurable task diverts effort from an unmeasurable one; Baker [2] shows that

adding a noisy measure of the true objective can *reduce* welfare. Our RQ1 result is a computational, heterogeneous-agent instantiation of these theorems—we concede it is not a new mechanism—while our RQ3 audit result speaks directly to Baker's noisy-measure concern with a concrete estimator-design remedy. The behavioral anchors for our response model are intrinsic-motivation crowding-out [4, 7] and the hygiene/motivator distinction [9]; we use these for the *sign and functional form* of mechanisms only, not for calibrated magnitudes.

Monitoring backfires. Non-monotone effects of monitoring on human effort are documented behaviorally via distrust and crowding-out [6, 5]. Our audit backfire, when it appears, arises from a different source—estimator variance in the learner, not worker psychology—and, crucially, we show it is removable by estimator design, which the behavioral backfires are not.

RL for retention. Prior applied work uses RL to *maximize* observed retention or customer value [3]; none, to our knowledge, characterizes the damage to an unobserved welfare construct or the estimator dependence of an audit remedy.

3 Model and Method

Workforce. We simulate N=400 employees over T=12 quarters. Each employee has one of four archetypes in equal share—low-compensation, low-career-velocity, burnout-prone, and ambition-driven—whose structure and archetype-to-incentive affinities extend a prior incentive taxonomy the author derived from a public HR-analytics dataset. Each employee carries a latent satisfaction $s \in [0, 1]$; *s is a unitless illustrative variable and all magnitudes below are non-calibrated modelling choices.*

Incentives and dynamics. Each quarter the allocator gives each employee one of five incentive categories (monetary, recognition, flexibility, development, social) or none, subject to a per-capita budget. An incentive matched to an archetype's affinity raises s with diminishing returns; untreated s drifts down; repeated identical incentives habituate. Monetary incentives additionally impose a one-quarter-lagged crowding-out penalty on the ambition and burnout archetypes [4, 7].

Attrition and the handcuff channel. An employee leaves with probability

$$\Pr(\text{leave}) = \sigma(\beta_0 - \beta_s\, s - h\mathbb{1}[\text{monetary in last two quarters}])\,, \tag{1}$$

where σ is the logistic function, β_0 and β_s are illustrative constants, and $h \geq 0$ is the *handcuff strength*: the degree to which recent monetary incentives suppress attrition *without* raising satisfaction (modelling vesting, switching costs, or non-competes). h is the wedge that decouples the observed proxy from the latent objective; we sweep it. Leavers are replaced by fresh employees (sensitivity to this choice is tested in §5).

Policies. We compare six allocators: *Random*; *Monetary-only*; *Static*, a theory-based rule assigning each archetype its affinity-matched category (the strongest non-learning baseline); *Learner*, a per-archetype ε-greedy bandit rewarded on retention; *Learner+audit*, which additionally surveys a random fraction p of employees each quarter and folds the audited satisfaction into its reward; and *Oracle*, which sees true s (an upper bound). The learner maintains two per-cell estimates and selects the action maximizing

$$v(k, a) = \hat{Q}^{\text{stay}}(k, a) + \lambda\, \phi\big(\hat{Q}^{\text{sat}}(k, a), n_{k,a}\big), \tag{2}$$

where k is the archetype, $\hat{Q}^{\text{stay}}$ and $\hat{Q}^{\text{sat}}$ are running means of retention and audited satisfaction, $n_{k,a}$ is the number of audited observations, λ weights satisfaction, and ϕ is the *satisfaction estimator*. We study three choices of ϕ: `raw` (the unshrunk mean at full weight), `shrink` (precision-weighted toward a neutral prior, weight $n/(n + K)$), and `gated` (used only once n exceeds a threshold). This estimator choice is the object of RQ3.

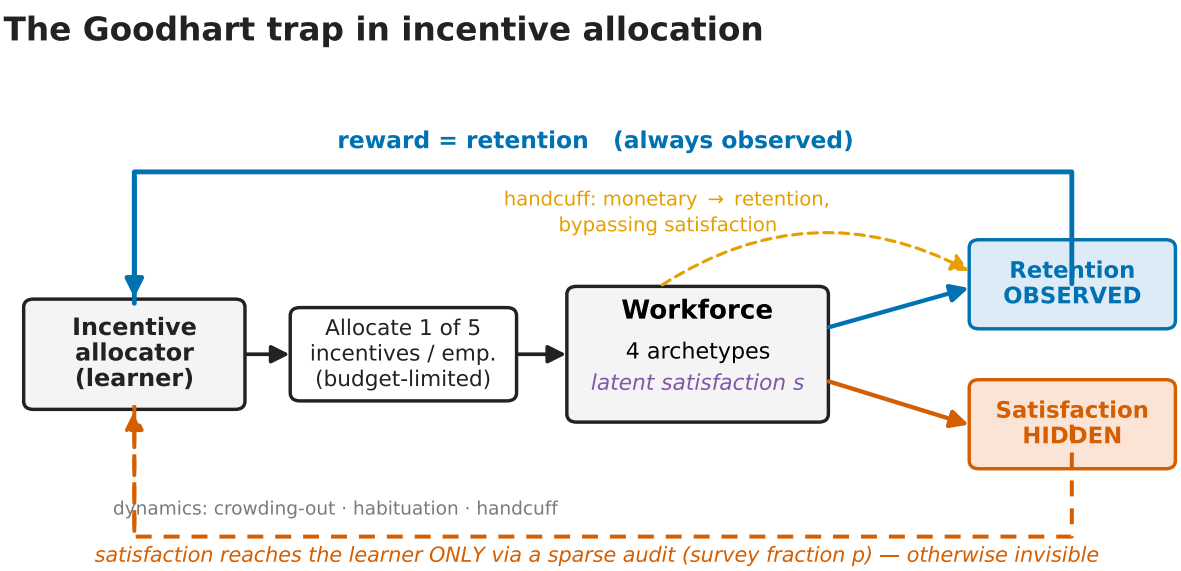


Figure 1: *(AI-generated schematic.)* The partial-observability incentive loop. The allocator's reward sees only retention; latent satisfaction returns only through a sparse audit of fraction p. The handcuff channel lets monetary incentives raise retention without raising satisfaction, decoupling the optimized proxy from the true objective.

4 Experimental Setup

We run every policy for 200 random seeds at each handcuff strength $h \in \{0, 0.5, 1, 1.5\}$; the audited learner additionally sweeps $p \in \{0, 0.05, 0.1, 0.2, 0.4\}$ and the three estimators. We report, per run, *observed retention* per quarter, *mean latent satisfaction* (unobserved by the allocator), the *archetype gap* (max−min mean satisfaction across archetypes; an equity measure), and, for the confound controls, mean satisfaction over the never-replaced *founder cohort*.

Statistics. Policy contrasts are paired by seed. We report absolute mean differences with 95% confidence intervals and, for every directional claim, the per-seed *sign-stability* (fraction of seeds agreeing on the sign). We deliberately do *not* headline standardized effect sizes: in a seed-controlled simulation they measure within-model separation and inflate with population size, not behavioral magnitude. Kill-test families are corrected with Benjamini–Hochberg FDR.

Validation. A pre-run battery confirms the learner recovers each archetype's optimal category under an aligned reward, the attrition hazard is monotone in s, the budget constraint is never violated, and runs are deterministic per seed.

Pre-registered kill-tests. Before the confirmatory runs we registered four falsification conditions, each targeting a way the headline could be an artifact: (K1) if the satisfaction gap vanishes at zero crowding-out, it is merely a set constant; (K2) if it vanishes on the founder cohort, it is survivorship composition; (K3) if the audit backfire vanishes under a precision-aware estimator, it is an estimator artifact; (K4) if signs flip across a randomized parameter sweep, the findings are not robust.

Table 1: Main configuration (h=1, 200 seeds, ± = 95% CI). Higher retention and satisfaction are better; lower archetype gap is more equitable. The Learner matches Static on the observed metric while degrading the two unobserved ones. Regenerates from `results/full.csv`.

Policy	Retention/qtr	Latent satisf.	Archetype gap	Monetary share
Random	0.938	0.541	0.034	0.16
Monetary-only	0.944	0.467	0.154	0.50
Static (theory)	0.941	**0.566**	**0.021**	0.19
Learner (retention)	0.941	0.529	0.066	0.25
Learner + audit (p=.1)	0.942	0.525	0.082	0.28
Oracle (true s)	0.943	0.542	0.069	0.25

5 Results

The learner degrades the objective it cannot see while matching the baseline on the one it can. At the main configuration (h=1, Table 1, Fig. 2), the Learner and the Static baseline are statistically indistinguishable on observed retention (0.941 vs. 0.941; paired difference −0.0005, sign-stability 54%), yet the Learner's latent satisfaction is far below (0.529 vs. 0.566; paired difference −0.037 [95% CI ± 0.002], 100% sign-stable) and its archetype inequality is roughly tripled (0.066 vs. 0.021). The gap on the observed metric even reverses sign with the handcuff channel: the Learner is *worse* than Static on retention at h=0 (+0.0061), tied at h=1, and *better* at h=1.5 (−0.0059)—it "wins" on the observable proxy precisely where that proxy is most decoupled from welfare.

The degradation is genuine harm, not survivorship composition, and not a hand-set constant. We registered two ways this headline could be an artifact and tested both (Fig. 3). At *zero* crowding-out and *zero* handcuff—every welfare-suppressing channel switched off—the Static−Learner satisfaction gap does not vanish but remains +0.0151 [±0.0010] (98% sign-stable), attributable to the learner's exploration cost and habituation; crowding-out and the handcuff then amplify it monotonically. Restricting the metric to the never-left *founder* cohort leaves the gap essentially unchanged (+0.0141 [±0.0011]), so it is per-person harm rather than the retention of already-unhappy workers; the gap is likewise robust to not replacing leavers (+0.024) and to drawing replacements from the survivor distribution (+0.049). K1 and K2 do not fire. We therefore read RQ1/RQ2 as a heterogeneous-agent instantiation of multitask agency [10]: a robust, decomposed, but theoretically anticipated effect.

The welfare and equity erosion survives calibration to a real benchmark. To test whether the effect is an artifact of illustrative parameters, we calibrated the two most load-bearing choices to the IBM HR Analytics benchmark (n=1470; Fig. 5): the satisfaction→attrition hazard slope, by logistic regression of attrition on job satisfaction, and the archetype mix, by clustering. The fitted slope is 0.75–0.95 (vs. our illustrative 3.2), the archetype mix is skewed (low-compensation 39% vs. an equal-share 25%), and—notably—an off-satisfaction retention channel is empirically present: a stock-option coefficient of −0.51 [−0.72, −0.31] controlling for satisfaction, grounding the "handcuff" construct in data. Re-running with the fitted slope and real mix, the welfare gap holds (+0.033 vs. +0.037 illustrative) and the equity gap is if anything larger (+0.047 vs. +0.045); both persist on the founder cohort. The distributional erosion is thus not a parameter artifact.

The audit remedy: an honest negative result. The natural remedy is to survey a fraction p of employees and reward the learner on measured satisfaction. We initially observed, with a naive `raw` estimator, a striking backfire—auditing at p=0.05 yielding *lower* satisfaction than no audit (−0.0085, $p_{\mathrm{FDR}}<10^{-13}$ at h=0)—as a low-precision estimate enters the argmax at full weight and flips allocations on noise. Two pre-registered tests dismantle the strong version of this claim (Fig. 4).

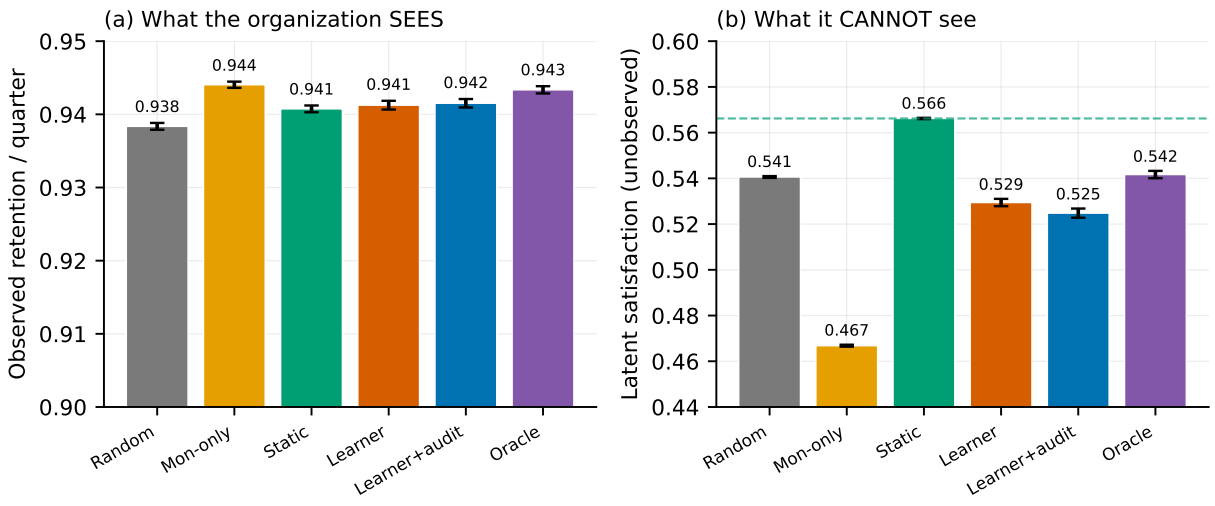


Figure 2: *(AI-generated.)* h=1, 200 seeds, 95% CI. **(a)** What the organization observes (retention): the Learner matches Static. **(b)** What it cannot see (latent satisfaction): the Learner falls well below Static (dashed line). The two panels use the same policies; only the measured axis differs.

First (K3), a precision-weighted `shrink` estimator removes the backfire entirely (non-significant after FDR at both h). Second, and more decisively, under the *calibrated* (weaker) hazard slope the backfire vanishes even for the `raw` estimator (+0.003): it required a strong satisfaction–retention coupling the benchmark does not exhibit. We therefore report a negative result: naive auditing backfires only in a narrow regime (strong coupling × naive estimator); under realistic coupling or competent estimation, auditing the hidden objective is roughly neutral-to-helpful. Audit design and the coupling regime, not audit frequency, determine whether measurement helps—and in the calibrated regime the remedy neither hurts nor much helps.

Robustness. Over 250 randomized worlds (gains, crowding, and hazard slope drawn over plausible ranges) the population and founder welfare-gap signs and the equity-gap sign hold in 100% of worlds, while the raw audit backfire holds in only 62% (Appendix Fig. 6)—quantifying which claims are robust (the welfare/equity erosion) and which are regime-specific (the naive backfire).

6 Discussion

Two implications follow. First, and robustly (across randomized worlds and under benchmark calibration), an allocator evaluated only on the metric it optimizes can look healthy while eroding an unmeasured objective and—more sharply—its *fair distribution* across worker types, looking *healthiest* exactly where the proxy is most decoupled from welfare. The distributional consequence, not just the average, is what an aggregate KPI hides. Second, and more soberly: the intuitive fix of occasionally measuring the true objective is, under realistic (benchmark-calibrated) conditions, roughly neutral—it neither reliably helps nor, once estimated with basic care, hurts. The naive-auditing backfire we first found was a fragile, strong-coupling phenomenon, not a general law; we report it as such. The actionable message is therefore about *measurement of distribution*, not a silver-bullet audit remedy: track per-cohort welfare, because the aggregate will not show you where the harm lands.

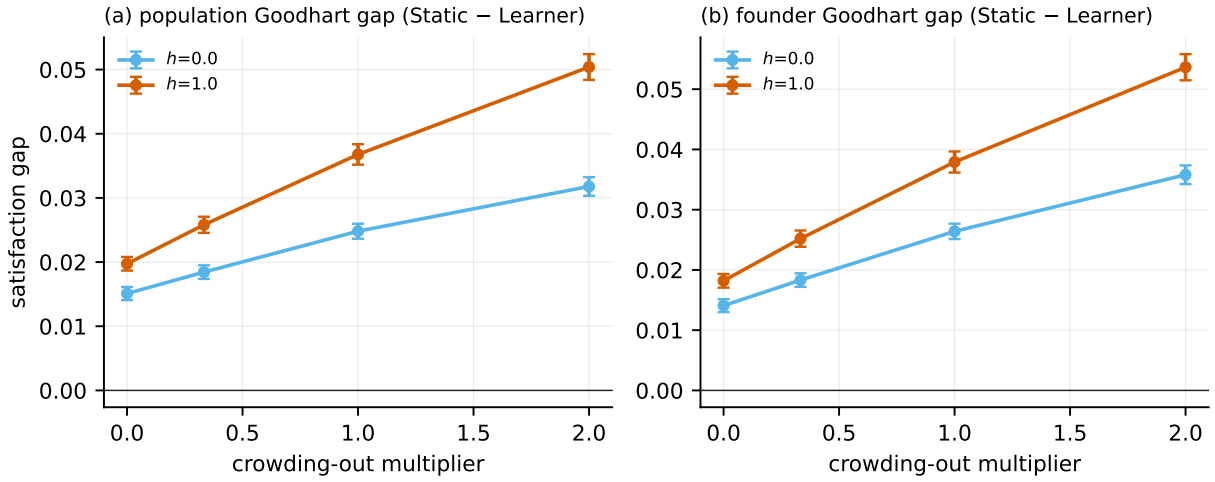


Figure 3: *(AI-generated.)* Static−Learner satisfaction gap vs. the crowding-out multiplier, for the full population (a) and the never-left founder cohort (b). The gap is nonzero with all amplifiers off (crowding = 0, and h=0, left points), then grows with crowding and with the handcuff h. The founder panel rules out survivorship composition. 200 seeds, 95% CI.

7 Limitations

Simulation realism. The study is a conditional model. We calibrate the two most load-bearing quantities (the satisfaction→attrition hazard slope and the archetype mix) to the IBM HR Analytics benchmark, but the benchmark is itself a synthetic dataset and its satisfaction is a cross-sectional self-report, so the satisfaction *dynamics* (gains, crowding, habituation) remain illustrative; magnitudes are structural, not field-estimated. The defense against "you assumed your conclusion" is the joint evidence of the randomized-world sweep, the zero-crowding control, and the benchmark calibration—not any single parameter. **Response-model family.** A single functional family is used for the dynamics; a follow-up should calibrate these to a longitudinal turnover-and-engagement panel, which our cross-sectional benchmark cannot supply. **Learner class.** The optimizer is a cohort ε-greedy bandit; a contextual learner might shift magnitudes, though a stronger optimizer on the same misaligned proxy should, if anything, degrade welfare more. **Welfare accounting.** We do not track post-exit outside options; "degradation" is a within-model latent, not a claim about lifetime worker welfare.

8 Conclusion

Optimizing an observable retention proxy degrades unobserved satisfaction and, more sharply, its fair distribution across worker types—an effect that survives a zero-crowding control, a founder-cohort test, a 250-world randomized sweep, and calibration of the hazard slope and archetype mix to a real HR benchmark, in which the off-satisfaction retention channel is itself empirically present. We frame this as a distributional sharpening of multitask agency rather than a new mechanism: the contribution is the calibrated, falsification-tested demonstration and the equity axis, not the aggregate Goodhart effect, which is anticipated. The intuitive remedy—auditing the hidden objective—we report honestly as a near-null under realistic conditions: the naive-auditing backfire we first found was a fragile, strong-coupling phenomenon that vanishes under competent estimation

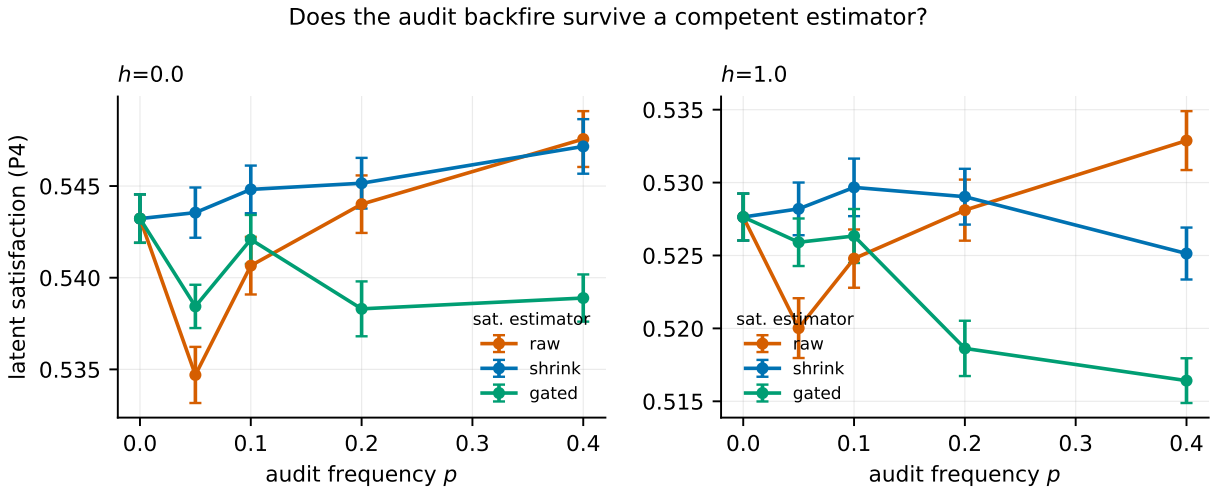


Figure 4: *(AI-generated.)* The audit backfire is an estimator artifact. Audited-learner satisfaction vs. audit frequency p under a naive running-mean (`raw`), a precision-weighted (`shrink`), and a confidence-gated (`gated`) satisfaction estimator. The low-p dip under `raw` is removed by `shrink` (non-significant after FDR at both h). 200 seeds, 95% CI.

or benchmark-calibrated coupling. The durable lesson is to measure the *distribution* of the true objective, not merely its average, because an aggregate proxy hides where the harm concentrates. We release the testbed, the pre-registered kill-tests, and the calibration so these findings can be pushed further toward field data.

Acknowledgments and disclosure

Simulation code, analysis code, and figure rendering were AI-assisted (a large language model). The research questions, experimental-design decisions, execution of the experiments of record, interpretation, and the intellectual content are the author's own; an AI-use prompt log is available on request. Adversarial review of an earlier draft (which surfaced the estimator-artifact hypothesis subsequently confirmed by kill-test K3) was AI-assisted and is likewise logged.

References

[1] Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., & Mané, D. (2016). Concrete Problems in AI Safety. *arXiv:1606.06565*.

[2] Baker, G. (1992). Incentive Contracts and Performance Measurement. *Journal of Political Economy*, 100(3), 598–614.

[3] Cheng, X. (2020). Obtain Employee Turnover Rate and Optimal Reduction Strategy Based on Neural Network and Reinforcement Learning. *arXiv:2012.00583*.

[4] Deci, E. L., Koestner, R., & Ryan, R. M. (1999). A meta-analytic review of experiments examining the effects of extrinsic rewards on intrinsic motivation. *Psychological Bulletin*, 125(6), 627–668.

[5] Dickinson, D., & Villeval, M.-C. (2008). Does monitoring decrease work effort? The complementarity between agency and crowding-out theories. *Games and Economic Behavior*, 63(1), 56–76.

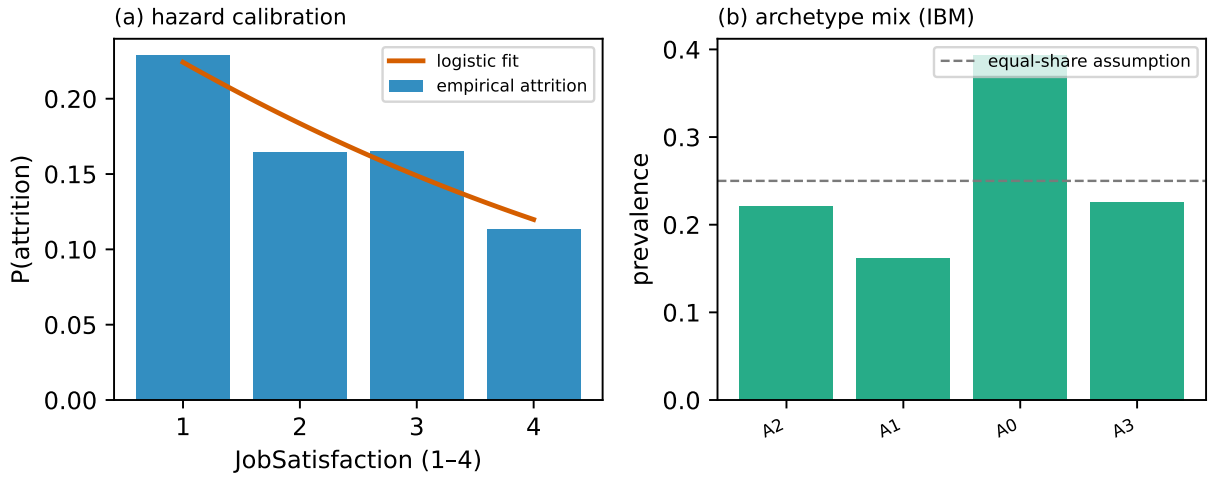


Figure 5: *(AI-generated.)* Calibration to the IBM HR Analytics benchmark (n=1470). **(a)** Empirical attrition falls monotonically with job satisfaction; logistic fit overlaid (slope 0.75–0.95, vs. our illustrative 3.2). **(b)** The archetype mix departs from the equal-share assumption (dashed): low-compensation employees are 39% of the workforce. The welfare/equity erosion holds under these fitted values.

[6] Falk, A., & Kosfeld, M. (2006). The Hidden Costs of Control. *American Economic Review*, 96(5), 1611–1630.

[7] Frey, B. S., & Jegen, R. (2001). Motivation Crowding Theory. *Journal of Economic Surveys*, 15(5), 589–611.

[8] Gao, L., Schulman, J., & Hilton, J. (2023). Scaling Laws for Reward Model Overoptimization. *ICML* (PMLR 202).

[9] Herzberg, F., Mausner, B., & Snyderman, B. B. (1959). *The Motivation to Work*. Wiley. [Two-factor theory; note it is contested in modern I/O psychology, e.g. House & Wigdor 1967.]

[10] Holmström, B., & Milgrom, P. (1991). Multitask Principal-Agent Analyses: Incentive Contracts, Asset Ownership, and Job Design. *Journal of Law, Economics, & Organization*, 7(Special Issue), 24–52.

[11] Karwowski, J., et al. (2024). Goodhart's Law in Reinforcement Learning. *ICLR*. arXiv:2310.09144.

[12] Krakovna, V., et al. (2020). Specification gaming: the flip side of AI ingenuity. *DeepMind Blog*.

[13] Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1), 68–78.

[14] Skalse, J., Howe, N. H. R., Krasheninnikov, D., & Krueger, D. (2022). Defining and Characterizing Reward Hacking. *NeurIPS*. arXiv:2209.13085.

[15] Strathern, M. (1997). 'Improving ratings': audit in the British University system. *European Review*, 5(3), 305–321.

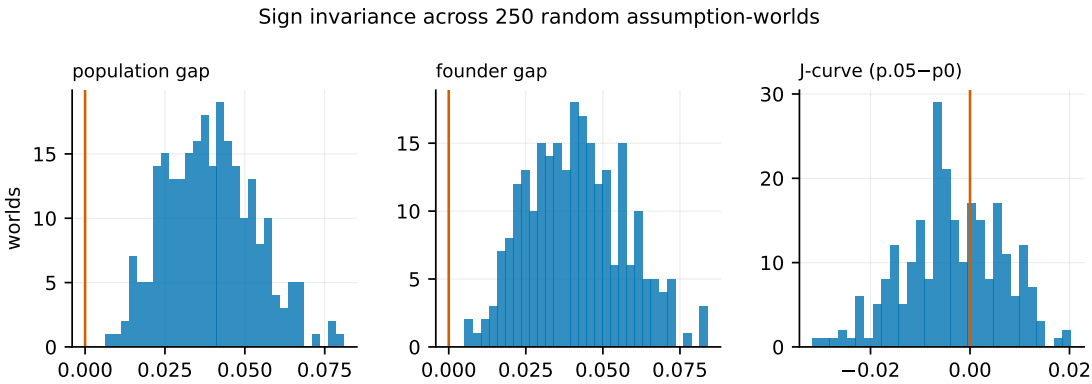


Figure 6: *(AI-generated.)* Sign invariance across 250 randomized assumption-worlds. Population and founder Goodhart-gap signs hold in 100% of worlds (mass entirely > 0); the raw-estimator audit backfire holds in 62% (mass straddles 0)—quantifying which findings are robust and which are not.